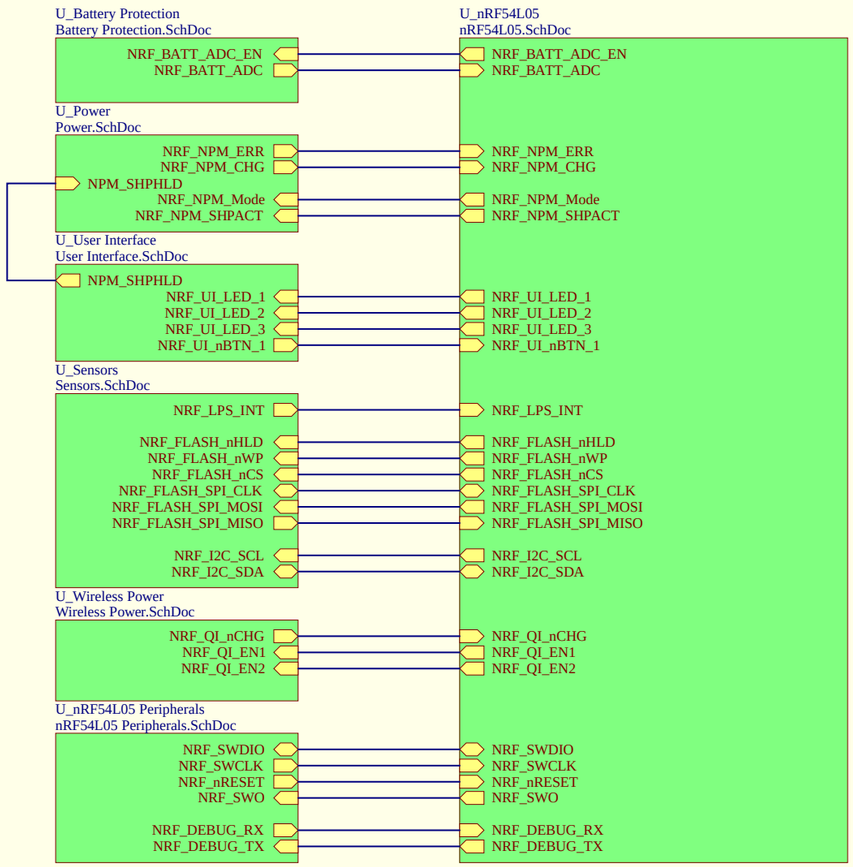
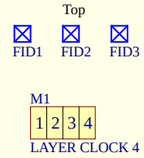
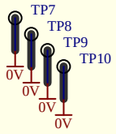


I2C Address Table

I2C0

Device	Part Number	Address (7bit)	Power Rail
Temperature and Relative	SHT40-AD1B-R3	0x44	3V0
CO2 Sensor	STCC4	0x64	3V0
Pressure Sensor	LPS22HHTR	0x5D	3V0







U_Revison History A3
Revison History A3.SchDoc

PWA Title:
CO2 Monitor

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

Sheet Title: **Top**

Size: A3	Number: AH-013-0001-SCH	Revision: 1A
Print Date: 31/10/2025 12:45:47 PM		Sheet 1 of 9
File: Top.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 29/10/2025

1S LiPo Protection IC

+VBATT

J1 to be used for power testing and battery isolation as battery will be directly soldered on

+VBATT_PROT

TVS on +VABTT_PROT output pin

J1
JUMPER 2PIN

F1
250mA / 32V
0467.250NR
Fuse: UL File No. E10480

R3
0R

Mechanically can not be attached backwards. Thus, no reverse polarity.

BATT1
153040 140mAh

R1
330R

C1
100n
16V

U1
BQ29700DSER

BAT
VSS
DOUT
COUT
V-
NC

Over charge detection = 4.350V
Over charge release = 4.150V
Over Discharge detection = 2.3V
Over Discharge relaese = 3.0V
Over current detection voltage = 0.2V

TP4
TP5
TP11

R2
2k2

Q1
PMF63UNE
Q2
PMF63UNE

0V

Rds = 50mohm each
System Imax = 1A
System Iavg = <50uA

MOSFET Vgs(ths) has to be greater than 400mV and less than 4.350V
Our MOSFET Vgs(ths) = 1V @ 25deg (min 500mV)
Vgs(thsmax) = 1V
Vgs(max) = 8V



Voltage Detection

+VBATT_PROT

R41
1M0

Q4B
DMC2400UV

Q4A
DMC2400UV

NRF_BATT_ADC_EN

R42
1M0

R43
1M0

NRF_BATT_ADC

R44
1M5

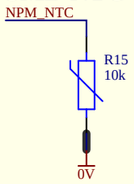
Scaling factor of 0.6.
Vbatt 4.2V = Vadc 2.52V
Vbatt 3.7V = Vadc 2.22V
Vbatt 3.3V = Vadc 1.98V

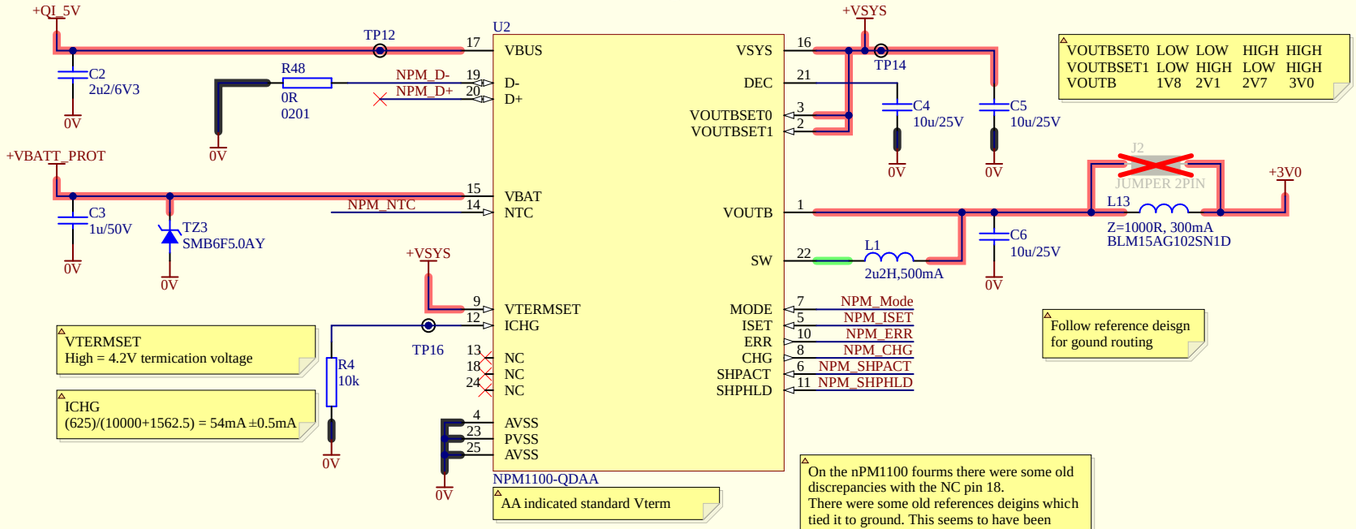
0V

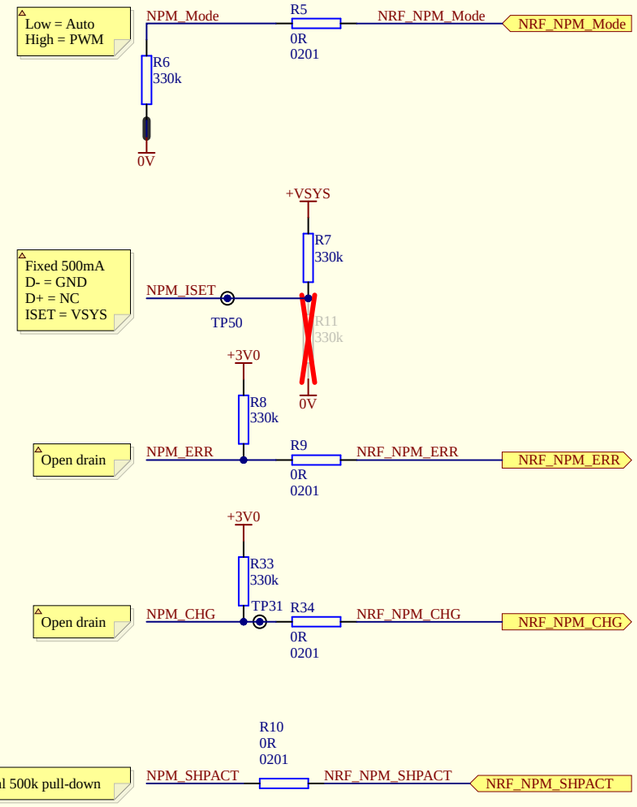
PWA Title:	CO2 Monitor
Genesys Electronics Design ABN 62 112 253 424 Unit 5, 33 Ryde Rd, Pymble NSW, 2073, Australia Ph: +61 2 9496 8900	Sheet Title: Battery Protection Size: A3 | Number: AH-013-0001-SCH | Revision: 1A Print Date: 31/10/2025 12:45:47 PM | Sheet 2 of 9 File: Battery Protection.SchDoc Drawn: G.T | Approved: G.S. | Approved Date: 29/10/2025

Power Management

SMD NTC







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PWA Title:
CO2 Monitor

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
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Sheet Title: **Power**

Size: A3	Number: AH-013-0001-SCH	Revision: 1A
Print Date: 31/10/2025 12:45:47 PM		Sheet 3 of 9
File: Power.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 29/10/2025

Wireless Power Reciever

Place close togeather
80kHz -3dB cutoff

U3
BQ51013BRHLR

6	COMM1	OUT	4
5	CLAMP1	RECT	18
3	BOOT1	FOD	14 QI_FOD
2	AC1	TS/CTRL	13 QI_TS
19	AC2	ILIM	12 QI_ILIM
17	BOOT2	AD	9
16	CLAMP2	CHG	7 QI_nCHG
15	COMM2	AD-EN	8 QI_nAD_EN
		EN1	10 QI_EN_1
		EN2	11 QI_EN_2
		PGND	1
		PGND	20
		PAD	

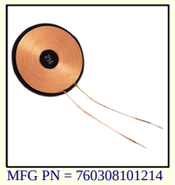
C7 47n
C8 22n
C9 470n/25V
C11 68n
C12 10n
C15 68n
C16 1n8
C17 1n8
C18 10n
C19 470n/25V
C20 22n
WP6
WP7

+QI_5V_RAW
C10 2u2/6V3
TZ2 SMB6F5.0AY
C13 10u/25V
L2 390nH,170mA 0603HP-R39XJLW
+QI_5V
C21 10u/25V
C22 10u/25V
C14 4u7/16V
QI_RECT

QI_nAD_EN leave floating

RX Coil

19mm Diameter x 0.8mm Height
H = 26µH
I = 1.1A, R = 240-520 mR
https://octopart.com/search?q=760308101214



MFG PN = 760308101214

SMD NTC

QI_TS
R18 10k
0V

Additional RTC for monitoring board temp while charging.
The main battery NTC lives on the power sheet

QI_RECT TP52
R13 20k / 0.1%
Adds option for tuning FOD
Currently using default values (tesed with default values tested with devkit)
QI_ILIM TP53
R14 2k7
QI_FOD TP54
R16 100R / 0.1%
R17 91R

ILIM
K=314
RLIM=314/ILIM

Iout(A)	R(ohm)
0.62	510
0.56	560
0.51	620
0.42	750
0.35	910
0.31	1,000
0.24	1,300
0.17	1,800
0.12	2,700
0.06	5,100

R30 0R 0201
QI_EN_1TP35
QI_EN_2TP36
NRF_QI_EN1
NRF_QI_EN2
R31 0R 0201

Internal 200k pull-down
0 0 = Enabled with handshake

+3V0
R19 330k
QI_nCHG
R20 0R 0201
NRF_QI_nCHG

5mA capable open-drain

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PWA Title: **CO2 Monitor**

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

Sheet Title: **Wireless Power**

Size: A3	Number: AH-013-0001-SCH		Revision: 1A
Print Date: 31/10/2025 12:45:47 PM			Sheet 4 of 9
File: Wireless Power.SchDoc			
Drawn: G.T	Approved: G.S.	Approved Date: 29/10/2025	

NRF_BATT_ADC_EN
NRF_BATT_ADC

NRF_NPM_Mode
NRF_NPM_ERR
NRF_NPM_CHG
NRF_NPM_SHPACT

NRF_QI_EN1
NRF_QI_EN2
NRF_QI_nCHG

NRF_LPS_INT

NRF_UI_LED_1
NRF_UI_LED_2
NRF_UI_LED_3
NRF_UI_nBTN_1

NRF_FLASH_nHLD
NRF_FLASH_nWP
NRF_FLASH_nCS
NRF_FLASH_SPI_CLK
NRF_FLASH_SPI_MOSI
NRF_FLASH_SPI_MISO

NRF_I2C_SCL
NRF_I2C_SDA

NRF_SWDIO
NRF_SWCLK
NRF_DEBUG_TX
NRF_DEBUG_RX
NRF_nRESET
NRF_SWO

nRF54 Pins

32.768kHz XAL
Can use external or configurable internal capacitors
Tested with development kit

X2 32.768kHz

R45 22R
R46 22R
R47 22R

U7B

NRF_QI_nCHG	TP18	23	P0.00 (Blue Pins Are Dedicated Clock Pins)
NRF_NPM_ERR	TP19	24	P0.01
NRF_LPS_INT	TP20	27	P0.02
NRF_I2C_SCL	TP21	28	P0.03/GRTC_PWM
NRF_I2C_SDA	TP22	29	P0.04/GRTC_CLKOUT32K
NRF_XAL_1		1	P1.00/XL1
NRF_XAL_2		2	P1.01/XL2
NRF_DEBUG_TX		3	P1.02/NFC1
NRF_DEBUG_RX		4	P.1.03/NFC2
NRF_BATT_ADC_EN	TP27	5	P1.04/ASO0/AIN0
NRF_BATT_ADC	TP28	6	P1.05/ASI0/RADIO6/AIN1
NRF_FLASH_nHLD		7	P1.06/ASO0/AIN2
NRF_FLASH_nWP		8	P1.07/ASI1/AIN3
NRF_NPM_CHG		9	P1.08/CLKOUTFAST/EXTREF
NRF_NPM_Mode	TP32	37	P1.09/ASO2/RADIO0
		38	P1.10/ASI2/RADIO1
		39	P1.11/ASO3/RADIO2/AIN4
NRF_QI_EN1		40	P1.12/ASI3/RADIO3/AIN5
NRF_QI_EN2		41	P1.13/RADIO4/AIN6
		42	P1.14/RADIO5/AIN7
		11	P2.00/SPIM_DCX/UARTE_RXD/FLPR.4/QSPI_D3
NRF_SPI_CLK	TP2	12	P2.01/SPIM_SCK/SPIS_SCK/FLPR.0/QSPI_SCK
NRF_SPI_MOSI		13	P2.02/SPIM_SDO/SPIS_SDO/UARTE_TXD/FLPR.1/QSPI_D0
NRF_NPM_SHPACT	TP41	14	P2.03/FLPR.3/QSPI_D2
NRF_FLASH_SPI_MISO	TP42	15	P2.04/SPIM_SDI/SPIS_SDI/UARTE_CTS/FLPR.2/QSPI_D1
NRF_IC_FLASH_nCS		16	P2.05/SPIM_CS/SPIS_CS/UARTE_RTS/FLPR.5/QSPI_CS
NRF_UI_LED_2	TP44	17	P2.06/FLPR.6/SPIM_SCK/SPIS_SCK/TRACECLK
NRF_SWO		18	P2.07/FLPR.7/TRACEDATA[0]/SWO/SPIM_DCX/UARTE_RXD
NRF_UI_LED_1	TP46	19	P2.08/FLPR.8/TRACEDATA[1]/SPIM_SDO/SPIS_SDO/UARTE_TXD
NRF_UI_LED_3	TP47	20	P2.09/FLPR.9/TRACEDATA[2]/SPIM_SDI/SPIS_SDI/UARTE_CTS
NRF_UI_nBTN_1	TP48	21	P2.10/FLPR.10/TRACEDATA[3]/SPIM_CS/SPIS_CS/UARTE_RTS

nRF54L05-QFAA

nRF54 Low Power Reccomendations

- MCU pins to be actively set to analog mode when not in use.
- Peripherials to be disabled when not in use.
- DCDC regulator to be used.
- Extenral crystal to be used.
- TX power of -8dBm reccomended
- Port 1 should be disabled most of the time
- X2 to power GRTC
- SAADC on as little as possible

nRF54 Power + Misc

+3V0
+nRF_VDD
L6
Z=1000R, 300mA
BLM15AG102SN1D
C26 10u/25V
C27 100n

U7A

VDD	47	10	VDD
VDD	48	22	VDD
		36	VDD
DCC	46		DCC
DECD	45		DECD
DECA	43		DECA
DECRF	33		
VSS	44		VSS
VSS	49		VSS
XC1		34	XC1
XC2		35	XC2
ANT		31	ANT
VSS_PA		32	VSS_PA
SWDCLK		26	NRF_SWCLK
SWDIO		25	NRF_SWDIO
nRESET		30	NRF_nRESET

nRF54L05-QFAA

L3
4u7H, 120mA
MLZ1608M4R7WT000
C31 2u2/6V3
L4
Z=120R, 200mA
BLM18AG121SN1D
C32 2u2/6V3
C33 10n
C34 2n2
C28 100n
C29 100n
C30 100n

32MHz XAL
Can use external or configurable internal capacitors

X1
32.0MHz

NRF_BLE_ANTENNA
To Pin49
To Pin32 ONLY

L8 2n7H,+/-0.1nH
L9 3n6H,+/-0.2nH
LQW15AN3N6C10D
L10 3n6H,+/-0.2nH
C38 1p5,+/-0.1p
C39 2p0,+/-0.1p
C40 0p3,+/-0.05p
ONLY ground at bottom layer

U9
B39242B9413K610

R29 1k0
C36 3n9

SWCLK: Internal pull-up
SWDIO: Internal pull-down
nRESET: Internal pull-up

Important note on C38:
- Grounding of C38 capacitor must ONLY be connected to Pin32 (VSS_PA) in top layer.
- Pin32 must ONLY be connected to Pin49 (VSS center pad) underneath the IC package

Important note on C39:
- Grounding of C39 capacitor MUST be isolated from ALL ground layers except the bottom ground layer.

BLE Antenna

Chip Antenna Matching Network
R26 0R
R39 0R
C35 3p9,+/-0.25pF
L5 2n7H,+/-0.1nH
CHIP_ANTENNA

Chip Antenna

ANT1
GND
GND
ANT3216A063R2400A

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
Ph: +61 2 9496 8900

PWA Title: **CO2 Monitor**

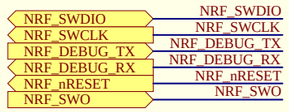
Sheet Title: **nRF54L05**

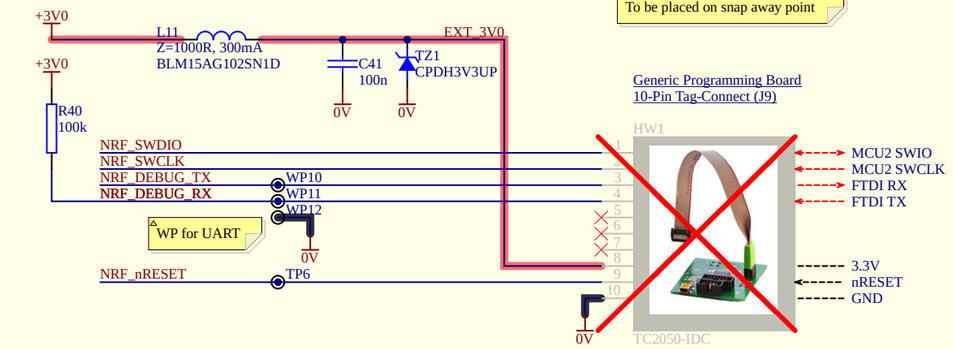
Size: A3	Number: AH-013-0001-SCH	Revision: 1A
Print Date: 31/10/2025 12:45:48 PM		Sheet 5 of 9
File: nRF54L05.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 29/10/2025

Programming & UART Debug

Ensure programmer does not supply 3V3 power.

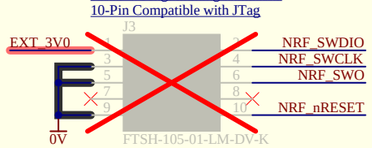
To be placed on snap away point

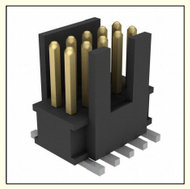




To be placed on bottom of board for inhouse development.
'Do not fit' we will only fit a few of these in house

Reserve Programming connector
10-Pin Compatible with JTag





PWA Title: **CO2 Monitor**		
Genesys Electronics Design ABN 62 112 253 424 Unit 5, 33 Ryde Rd, Pymble NSW, 2073, Australia Ph: +61 2 9496 8900	Sheet Title: **nRF54L05 Peripherals**	
	Size: A3 | Number: AH-013-0001-SCH	Revision: 1A
	Print Date: 31/10/2025 12:45:48 PM	Sheet 6 of 9
	File: nRF54L05 Peripherals.SchDoc	
	Drawn: G.T | Approved: G.S.	Approved Date: 29/10/2025

Temp & Humidity Sensor

I2C Address 0x44 (7bit)

Vcc = 1.08 V to 3.6 V
Operating Current = 2.2 uA (avg)
Idle Current = 1 uA
Temp. Range = -40C to 125C

Serial FLASH Memory

FLASH used for storing information required by the unit's software, including graphics images and logged data.

CO2 Sensor

I2C Address 0x64 (7bit)

Vcc = 2.7 V to 5.5 V
Operating Current = 500 uA (avg 10s)
Sleep Current = 1 uA
Temp. Range = 0C to 40C

Pressure Sensor

I2C Address 0x5D (7bit)

Vcc = 2.7 V to 5.5 V
Operating Current = 4 uA (avg)
Sleep Current = 0.9 uA
Temp. Range = -40C to 85C

Set to push-pull in firmware

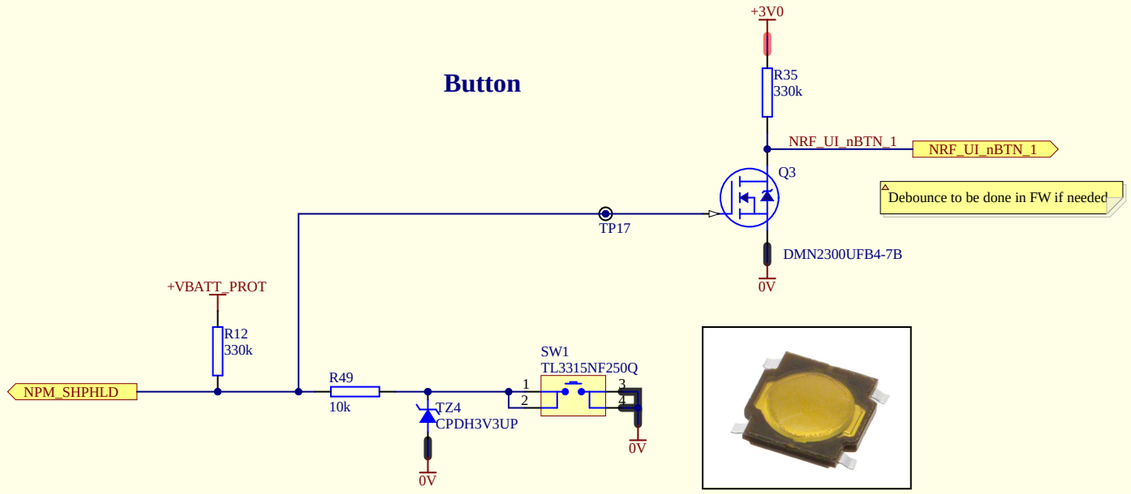
PWA Title: **CO2 Monitor**

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
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Sheet Title: **Sensors**

Size: A3	Number: AH-013-0001-SCH	Revision: 1A
Print Date: 31/10/2025 12:45:48 PM		Sheet 7 of 9
File: Sensors.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 29/10/2025

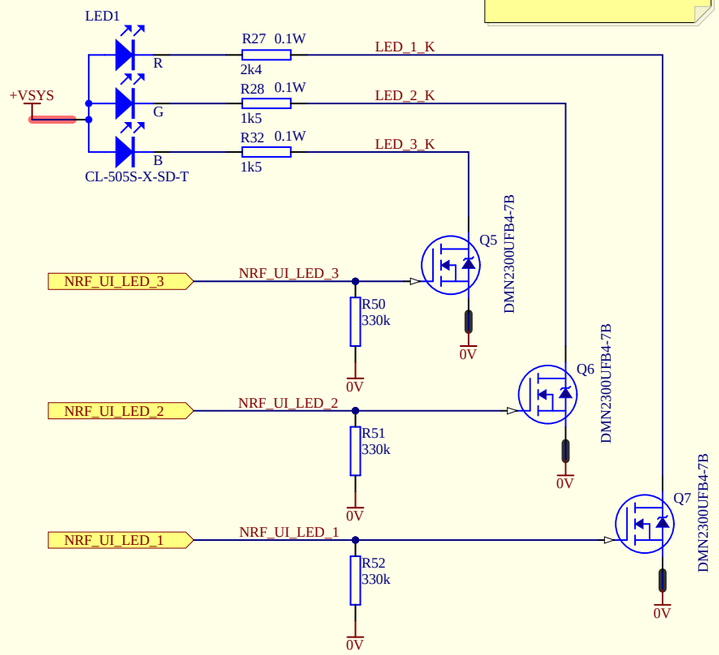
Button



LED Indicator

LED Drive Current = 5 mA (typ)
LED Pin Voltage = 0.5 V to VSYS

CL-505S-X-SD-T:
Ired = Vf = 1.85, 1mA @4V2
Igreen = Vf = 2.70, 1mA @4V2
Iblue = Vf = 2.75, 1mA @4V2



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PWA Title:
CO2 Monitor

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Sheet Title:
User Interface

Size: A3	Number: AH-013-0001-SCH	Revision: 1A
Print Date: 31/10/2025 12:45:48 PM		Sheet 8 of 9
File: User Interface.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 29/10/2025

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REV	CHANGES	DATE	ENG	APP
1A	Initial Release	29-Oct-25	GT	GS

REV	CHANGES	DATE	ENG	APP

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PWA Title:
Standard Build

Genesys Electronics Design
ABN 62 112 253 424
Unit 5, 33 Ryde Rd, Pymble
NSW, 2073, Australia
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Sheet Title:
Revision History

Size: A3	Master: AH-013-0001-SCH	Revision: 1A
Print Date: 31/10/2025 12:45:48 PM		Sheet 9 of 9
File: Revison History A3.SchDoc		
Drawn: G.T	Approved: G.S.	Approved Date: 29/10/2025

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